Round 17 | A full-tree quartic carrier

One explicit candidate; exact contraction algebra; accuracy gate still pending
Research response | 9 October 2026

I cannot substantiate a carrier meeting the requested correlation or cost. The candidate below preserves all index sectors and full birth history, but its reference readout bill is **649.014 u [D from M]**, before integration overhead. Its correlation is **unmeasured**. The attached script tests its untruncated, oracle-input version. I do not call it the cheapest passing carrier, or infer a universal cost lower bound.

Labels: [D] derived; [M] measured here on synthetic algebra/cost/software checks; [Q] quoted from your prompt or project code; [E] estimate. All equation constants, indices, dimensions and operation counts in formulas are [D] unless stated otherwise. No public-network accuracy was measured for this response.

The carrier: retain every connected-tree innovation

At pre-activation layer b, use the chain's mean μ, covariance C and *pre-ReLU* third cumulant K. Define $\sigma^2 = \mathrm{diag}(C)$, $\alpha = \mu/\sigma$ and the Gaussian derivative averages below. For a column query w in post-ReLU coordinates, define one quartic polynomial B_b:

```
d = Phi(alpha);  e = phi(alpha)/sigma
f = -alpha*phi(alpha)/sigma^2;  D = diag(d)
x = D w;  a = C x;  t = e * w * a
B_b[w^4] = 4 sum_i f_i w_i a_i^3
         + 12 t.T C t + 12 K[t,x,x]
```

The terms are Gaussian star, Gaussian path, and the full mixed K×C tree. Here * is elementwise multiplication. Use **full C**, including its diagonal, and all of K. Set K = 0 at the first Gaussian pre-layer. Start with no incoming fourth cumulant; append this innovation after each ReLU.

The mixed tensor has coefficient **12 Sym[$e_i d_j d_k d_l C_{ij} K_{ikl}$]**, where Sym averages index permutations. This creates both (2,1,1) and (1,1,1,1) entries. The path and star also populate these sectors. No entry is erased merely because its indices differ.

Transport and what is actually approximated

Use the prompt's row convention: W_b maps post layer b to pre layer b+1. A query at pre layer l pulls back through the following n by n propagator:

```
P_(b,b+1) = W_b.T
P_(b,l+1) = W_l.T D_l P_(b,l)
K4hat_l[v^4] = sum_(b<l) B_b[(P_(b,l).T v)^4]
```

This is exact linear transport of the stated approximate birth polynomial. It is not the exact ReLU cumulant update. It omits connected loops, nonlinear responses involving incoming K4, higher cumulants, and exact repeated-index resummation. The quoted one-step full-tensor oracle, 0.77-0.80 [Q], does not establish this carrier's accuracy. This distinction is why a ceiling test comes first.

Parity correction [D]: with upstream state fixed, a quartic contraction is even under global W → -W. Individual-coordinate sign structure remains. The third-cumulant odd-parity argument does not transfer to K4. Likewise, failure of the tested slice states [Q] is not a theorem about every matrix factorization.

State and exact diagonal readout

Reuse linear legs; freeze birth-specific coefficients

Keep, for each birth b, C_b [n,n], d_b, e_b, f_b [n]. Its Gaussian legs are P = $P_{b,l}$ and A = P $D_b C_b$ [n,n]. The latter equals the existing Gaussian off-diagonal A leg plus P diag(d_b diag C_b), when the same uncompressed transport is used. Define:

```
H = P * A;  J = H diag(e_b)
q_star = 4 rowsum(H * A^2 diag(f_b))
q_path = 12 rowsum((J C_b) * J)
```

For a symmetric cubic block Sym(X tensor Y tensor Z), freeze X_b, Y_b, Z_b at quartic birth b. The corresponding live legs are X_l = P $D_b X_b$, and similarly for Y,Z. Its mixed contribution is:

```
q_mix = 4 rowsum[(J X_b)*Y_l*Z_l
                 + (J Y_b)*X_l*Z_l
                 + (J Z_b)*X_l*Y_l]
```

Use the project's structure to avoid redundant dense factors

For cubic parent s, the source reconstruction in `r7_srcgain.py` [Q] is Sym(X tensor P tensor Y) + Sym(M tensor P tensor P) + Sym(V tensor P tensor y). In this paragraph P,A are the *parent's* current legs; scalars w,s,e,c1,c2 and right factors L,R1,R2 are fixed at its own birth. y is repeated over hub columns when needed.

```
X = 3A + Zf1 R1.T;  Y = A diag(w) + Zf2 R2.T
M = P diag(s) + 3A diag(e) + Z L.T
V = A diag(c1) + P diag(c2);  y = Z[:,-1]
F_A = 3P*Y + X*P*w + 3P^2*e + P*y*c1
F_P = X*Y + P^2*s + 2M*P + P*y*c2 + V*y
F_Z = P^2 L;  F_Z[:,-1] += rowsum(V*P)
F_Zf = [(P*Y) R1 | (X*P) R2]
```

Freeze only $B_{b,s}$ = [$A_{b,s}$ | $P_{b,s}$ | $Z_{b,s}$ | $Zf_{b,s}$] for every parent s<b. Cache its current cofactors F once per target layer. Then contract 4 rowsum[(J $B_{b,s}$)*F_s] and sum over parents. This uses **two square products plus one [n,n] by [n,22] product** per birth-parent pair [D; thin width from Q]. Z has 18 columns and Zf has 4 [Q]. The self-contained script implements and algebra-tests these cofactors.

Stored history [D]: full history has 105 frozen parent snapshots through the final birth, plus 15 local covariances/scalar sets: **909.20 MiB** in float32. Six active births at the last layer retain 69 snapshots plus local data: **582.00 MiB**. Neither includes live baseline legs, static cubic right factors or working buffers. Snapshots do not rotate; live legs follow U^T. Shared-basis projection changes this exact identity and needs a separate accuracy test.

Why this narrowly revisits a killed family

The local `r8_k4_memory_e2e.py` carries star terms; `r8_k4_dist_e2e.py` reads fresh K×C terms with restricted source age; the Q22 precheck uses per-source scalar weights [Q]. Here the full path and mixed innovations persist independently across all later layers. The frozen $B_{b,s}$ matrices are the extra quartic memory absent from those tests. This difference justifies the test, not a predicted pass.

Real-shape cost: far outside the envelope

n = 1024 [Q]; 1 u = 2^{31} FLOPs [Q]. Full-width kernel measurements; no live-row pruning.

I measured the project's fused Strassen class at recursion level 5 [Q], with flopscope 0.12.1+np2.2.6 [M]. Two dense mixed products share their left operand in one batch; the Gaussian product has a separate pool. Kernel charges below include scalar work and contractions; repeated calls use warm scratch. These are **exact meter observations for this implementation**, not asymptotic estimates.

Kernel [M]	Billed FLOPs	u	Ops
Gaussian star + path / birth	1,203,425,280	0.560389	125
Mixed / birth-parent pair	2,403,437,568	1.119188	118
Parent cofactors / layer	132,124,672	0.061525	49
Freeze one parent basis	2,119,680	0.000987	2
Freeze one local birth	1,053,696	0.000491	4

Let g_l be active quartic births and m_l their total cubic-parent count. All-history readouts have $g_l=l-1$ and $m_l=(l-1)(l-2)/2$. At six-age horizon, $g_l=6$ and $m_l=6l-27$ for $l\geq 8$ [D]. The following warm layer bill includes accumulation and normalization, but not birth snapshots or initial scratch:

```
F_l = 1,203,426,304 g_l + 2,403,438,592 m_l
      + 132,124,672(l-2) + 3,073   FLOPs
O_l = 126 g_l + 119 m_l + 49(l-2) + 5
```

Layer	Full: g / m	Full u / ops [D]	Six-age u / ops [D]
8	7 / 21	27.795 / 3,680	27.234 / 3,554
12	11 / 55	68.335 / 8,426	54.341 / 6,606
15	14 / 91	110.491 / 13,235	74.671 / 8,895
16	15 / 105	126.782 / 15,076	81.448 / 9,658

For mean-closure readouts at L8-L16 [schedule D; motivation Q], full history uses 99 Gaussian and 525 mixed reads; six-age uses 54 and 405 [D]. Both reference schedules take 90 parent-cofactor builds, 105 snapshots and 15 local births [D]. A birth at b costs 1,053,696 + (b-1)·2,119,680 FLOPs and 4+2(b-1) ops [D from M].

Network reference bill [D from M]	Full history	Six-age
FLOPs, including initial pools	1,393,746,443,273	1,051,179,628,553
Metered operations	79,779	59,829
u, including initial pools	649.013763	489.493659

Scope of the bill: additional readout/snapshot work with dense chain legs available for reuse. It excludes reconstructing old shared-basis legs, changed feedback, and integration overhead. It is not a measured whole-estimator total. Even this reference work exceeds +30 u / +1,000 ops [Q]. Classical products cost 1,166.881 u / 9,789 ops [D from M] for full history. More batching may lower operation count; none of these figures proves optimality.

The two Strassen scratch pools occupy 619.281 MiB [M]. Frozen full-history state plus these pools is 1.493 GiB [D], before live legs and other buffers; the whole-process 8 GB cap [Q] was not validated. No age, rank or sector truncation is currently certified to preserve correlation ≥0.70 [Q].

One falsifier, ready for your data

Round_17_Mixed_Tree_Ceiling.py | standalone NumPy; optional PyTorch CUDA

Primary test: all births, all cubic components, no rank projection or age cutoff. The script uses stored pre-teacher μ,C for coefficients and estimates the complete K3 directional contraction from new true-network samples. This is an **oracle-lower-moment test of the birth/transport closure**, not an exact true-K4 oracle and not a deployment simulation. A pass must subsequently survive chain-supplied inputs.

For each target l and birth b, pull all output-coordinate queries back as columns of $V = P_{b,l}^{T}$. Compute $X=D_bV$, $A=C_bX$ and $T=e_b[:,None]*V*A$. For samples δz_b, set $a=\delta z_b@X$ and $c=\delta z_b@T$. Then K3[T,X,X] is the third mixed cumulant κ(c,a,a), independently for each query column:

```
k3hat(c,a,a) = N^2/((N-1)(N-2)) *
 [mean(c*a^2) - mean(c)*mean(a^2)
  - 2 mean(a)*mean(a*c) + 2 mean(c)*mean(a)^2]
```

This contraction accesses the whole sampled K3 without constructing its tensor or fitting it to the teacher K4. The first pre-layer is exactly Gaussian, so its mixed term is zero. Stored K4d is read only for scoring. Baseline L{l}_ck4 sets the overall mean level of the prediction, without teacher-pattern fitting.

Run once from the project root

```
python "Architecture Analysis/Round_17_Mixed_Tree_Ceiling.py" \
  --root . --device auto --out round17_tree_ceiling_results.json
```

PowerShell users should put the command on one line. `--device auto` chooses CUDA when available, otherwise NumPy. Use `--device cpu --threads 8` for PyTorch CPU (thread count is a user setting). The diagnostic is outside the metered estimator. It imports no project Python modules and reads the public pre-teachers, weights and baseline dump at the paths in your prompt.

Pre-registered decision and runtime

Default $N = 2\times10^6$ [Q]; forwards and projections are float32, but **moment products and accumulation are float64**. Correlate predicted and stored-teacher excess kurtosis, each divided by true variance squared, on $|\alpha_{true}|<1.5$ [Q]. Print L12/L15/L16 [Q]: **PASS if every ρ≥0.70; KILL if every ρ<0.55; otherwise INCONCLUSIVE** [Q]. A secondary six-age readout shares the same samples. Half-sample correlations and a sampled-true-K4 control expose Monte Carlo instability; they do not silently change the gate.

There are 37 sampled birth-target mixed queries and 16 forward products per sample batch [D]. Dominant sampling work is 90Nn(2n-1) = 377.30 TFLOPs [D]; finishing within 2 hours [Q] requires at least 52.4 GFLOP/s sustained [D], plus float64 reductions and overhead. No hardware timing guarantee is established. The script prints an ETA [E] and stops between batches after its default 7,200-second limit [D setting]. Incomplete runs receive no PASS/KILL verdict.

What has been checked, and what has not

Synthetic dense-quartic identity: relative error 1.19×10^{-15} [M]. Optimized source-cofactor identity: 4.29×10^{-16} [M]. NumPy and Torch CPU completed a width-8, 70-sample fixture including an uneven batch split; partial-run gating also passed [M]. **No full public-network MC run, CUDA benchmark, correlation pass, or final-MSE improvement is claimed.** The PDF embeds the same script plus the meter log, meter probe and software-check log. Run the ceiling before prototyping compression.

Source trail: your Round 17 prompt [Q]; local `experiments/probes/r7_srcgain.py` (source factors), `submissions/v55k4c/estimator.py` (real shapes and Strassen implementation), and the narrow-kill probes named on page 2 [Q]. Primary background: ARC factor_k4.py, a symmetric pair-factor representation with linear transport. The polynomial, optimized cofactors and reported kernel measurements here are independently derived/tested; that reference is not evidence for their accuracy.